Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA: 3.18)

Academic Year:2026-27 SAP ID: 60003240262

DEPARTMENT OF INFORMATION TECHNOLOGY

COURSE CODE: DJS23ICPC501 **DATE:** 26/07/2026

COURSE NAME: Computer Networks **CLASS: T. Y. B. Tech**

EXPERIMENT NO. 1

CO/LO: Demonstrate networking commands.

AIM: Study of Linux Networking Commands

EXERCISE

Q. 1. Try the following networking commands and discuss the use and importance of each.

1. hostname

```
djsce-student@djsce-student-HP-ProOne-600-G6-22-All-in-One-PC:~$ hostname
djsce-student-HP-ProOne-600-G6-22-All-in-One-PC
```

The hostname command displays the name of the current computer (host) on the network. It is useful for identifying a device in local or enterprise networks and is often used in network administration and troubleshooting.

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2. nslookup

```
djsce-student@djsce-student-HP-ProOne-600-G6-22-All-in-One-PC:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 216.58.203.14
Name:   google.com
Address: 2404:6800:4009:808::200e
```

The nslookup command is used to query the Domain Name System (DNS). It converts domain names (e.g., google.com) into IP addresses and can also perform reverse lookups. It is commonly used to verify DNS records and diagnose DNS-related issues.

3. whois

```
djsce-student@djsce-student-HP-ProOne-600-G6-22-All-in-One-PC: $ whois google.com
   Domain Name: GOOGLE.COM
   Registry Domain ID: 2138514_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.markmonitor.com
   Registrar URL: http://www.markmonitor.com
   Updated Date: 2019-09-09T15:39:04Z
   Creation Date: 1997-09-15T04:00:00Z
   Registry Expiry Date: 2028-09-14T04:00:00Z
   Registrar: MarkMonitor Inc.
   Registrar IANA ID: 292
   Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
   Registrar Abuse Contact Phone: +1.2086851750
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
   Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
   Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
   Name Server: NS1.GOOGLE.COM
   Name Server: NS2.GOOGLE.COM
   Name Server: NS3.GOOGLE.COM
   Name Server: NS4.GOOGLE.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-08-07T10:02:31Z <<<

For more information on Whois status codes, please visit https://icann.org/epp
```

The whois command retrieves registration details of a domain name or IP address from WHOIS databases. It provides information such as the registrar, registration and expiry dates, and sometimes the owner's contact details (if publicly available). It is useful for domain management and cybersecurity investigations.

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4. route

```
djsce-student@djsce-student-HP-ProOne-600-G6-22-All-in-One-PC:~$ route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         10.120.64.1     0.0.0.0         UG    100    0        0 eno1
10.120.64.0     0.0.0.0         255.255.255.0   U     100    0        0 eno1
```

The route command displays or modifies the system's IP routing table. It shows how network packets are directed to their destinations and can be used to add, delete, or change routing entries for network communication.

5. arp

```
djsce-student@djsce-student-HP-ProOne-600-G6-22-All-in-One-PC:~$ arp -a
_gateway (10.120.64.1) at 00:87:31:c8:2a:eb [ether] on eno1
? (10.120.64.115) at b8:db:ad:ba:db:ad [ether] on eno1
```

The arp (Address Resolution Protocol) command displays and manages the ARP cache, which maps IP addresses to MAC addresses on a local network. It is useful for troubleshooting local network connectivity and verifying device communication.

6. IP

```
djsce-student@djsce-student-HP-ProOne-600-G6-22-All-in-One-PC:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 48:9e:bd:17:64:d5 brd ff:ff:ff:ff:ff:ff
    altname enp0s31f6
    inet 10.120.64.71/24 brd 10.120.64.255 scope global noprefixroute eno1
       valid_lft forever preferred_lft forever
    inet6 fe80::35c0:7f0a:f614:894a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: wlp0s20f3: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 04:cf:4b:6f:f3:15 brd ff:ff:ff:ff:ff:ff
```

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The ip command is a modern networking utility in Linux used to configure and display network interfaces, IP addresses, routing tables, and other networking information. It replaces older commands like ifconfig, route, and arp, providing a more comprehensive way to manage network settings.

CONCLUSION

The study of networking commands provides a practical understanding of how systems communicate and interact over a network. By learning these commands, one can efficiently configure network settings, monitor connectivity, troubleshoot issues, and gather essential diagnostic information, which is crucial for effective system and network administration.